

Climate 5000 Series - Gen 4

48K BTU Single Zone Regular Light Commercial System

Ducted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

48K Btu Single Zone Light Commercial Regular System - Ducted IDU

Indoor part number	8-733-962-714
Indoor model number	BMS500-AAU048-1AHDXD
Outdoor part number	8-733-962-693
Outdoor model number	BMS500-AAS048-1CSXLD

Efficiency Data AHRI 210/240 2023

SEER2	17.1
EER2	10.0
HSPF2 Region IV	9.8
HSPF2 Region V	7.7
COP(47F)	3.49
COP (5F)	2.00

Performance Data AHRI 210/240 2023

Cooling	Capacity Range	11000~56500
	Rated Capacity	48000
	Current	21.3
	Input Power	4800
Heating	Capacity Range	11200~60500
	Rated Capacity (47F)	50000
	Rated Capacity (5F)	38000
	Current	19.0
	Input Power	4200

Electrical Data

Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	5.0 / 39.0
Max. fuse indoor/outdoor (A)	15.0 / 40.0

Compressor Data

Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 1400
Input (W)	3700
Rated current (RLA/A)	7.0

Piping

Refrigerant precharge (oz/kg)	126.99 / 3.60
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	3/8" / Ø9.52
Gas side (in/mm)	3/4" / Ø19.0
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft/m)	246.0 / 75
Max. difference in height (ft/m)	98.4 / 30

1 of 8

Operation Range

ODU	Cooling (°F / °C)	-13 ~ 122 / -25 ~ 50
	Heating (°F / °C)	-13 ~ 75 / -25 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		8.5
IDU	Fan motor model	ZKFN-560-8-4
	Fan motor qty (ea)	1
	Fan motor input (W)	665.0
	Fan speed, H / M / L (RPM)	1000 / 950 / 880
	Air flow, Turbo / H / M / L / Si (CFM)	1648.1 / 1471.5 / 1294.9 / 1059.5 / N/A
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 51 / 46 / 42 / 40
	Net weight (lbs / kg)	111.33 / 50.5
	Gross weight (lbs / kg)	124.78 / 56.6
ODU	ESP range (Pa)	0-200
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-85-10-1L
	Fan motor qty (ea)	2
	Fan motor input (W)	126
	Fan motor speed, H / / L (RPM)	850 / 750 / 650
	Air flow, Max (CFM)	4237.9
	Noise level, Max (dB(A))	64.0
	Net weight (lbs / kg)	208.78 / 94.7
	Gross weight (lbs / kg)	239.86 / 108.8

Accessories

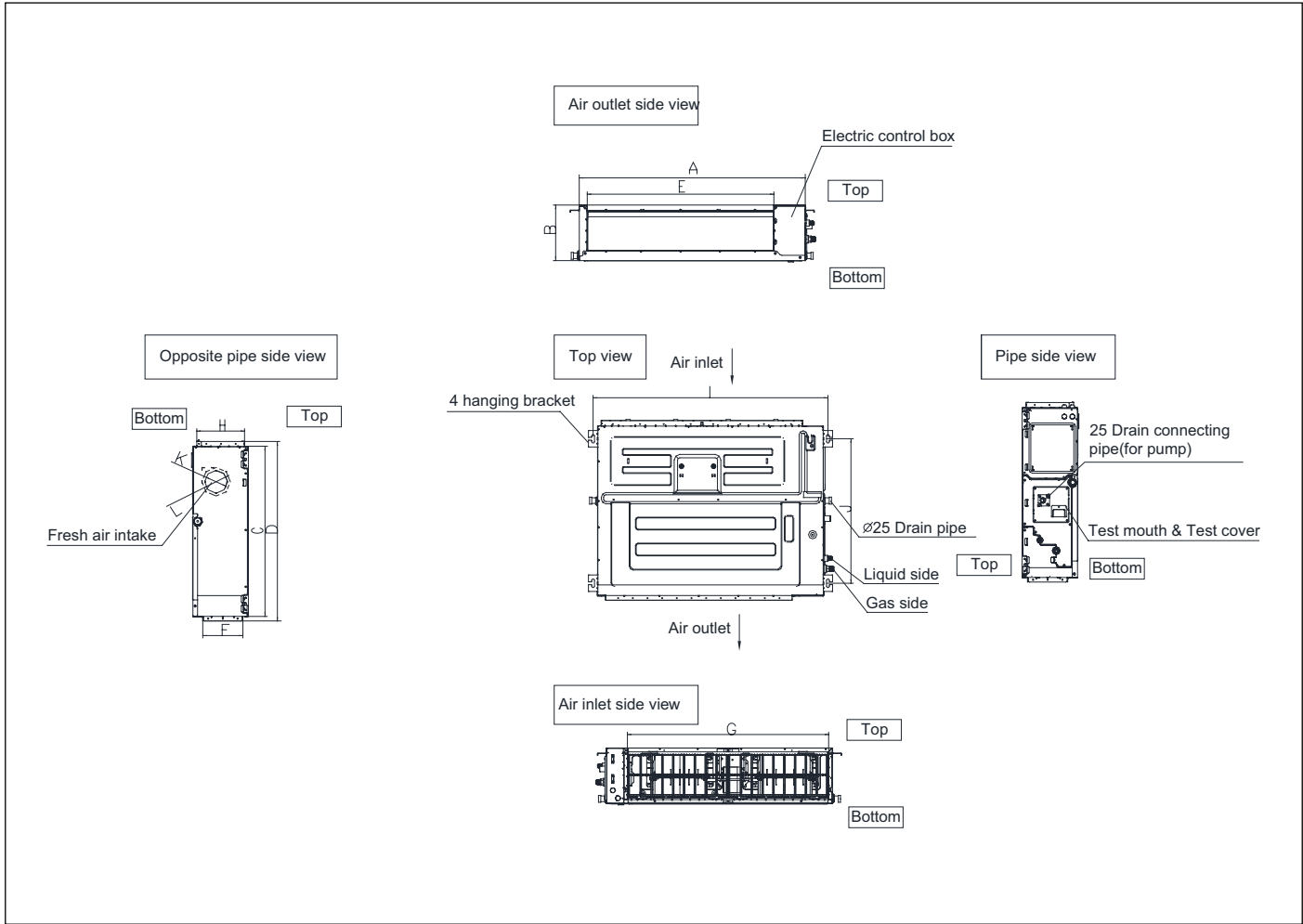
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface

Bosch Thermotechnology Corp.
 Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

Bosch Thermotechnology Corp. reserves the right to make changes without notice due to continuing engineering and technological advances | BTC 769208361 C | 01.2026

Tel: 1-800-283-3787 www.bosch-homecomfort.us

Indoor Unit Dimensions and Connections

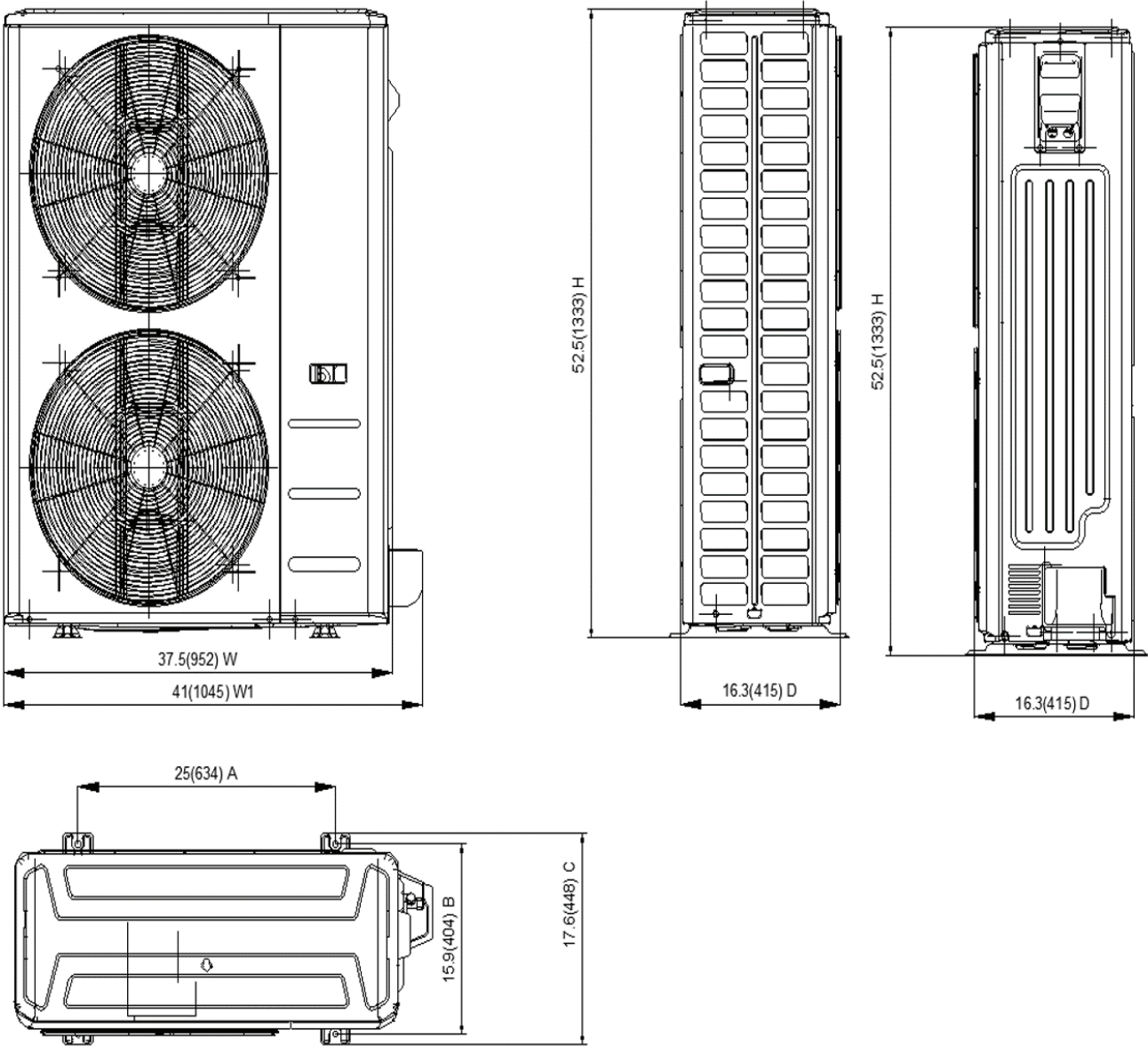


Model	Outline Dimension (in)				Air Outlet Opening Size (in)		Air Return Opening Size (in)		Size of Mounted Lug (in)		Fresh Air Intake Opening Size (in)	
	A	B	C	D	E	F	G	H	I	J	K	L
BMS500-AAU048-1AHDXD	47.2	11.8	29.5	31.3	40.4	9.2	43.0	10.5	48.8	25.2	4.9	6.3

Climate 5000 Series - Gen 4
48K Btu Single Zone Regular Light Commercial System - Ducted IDU

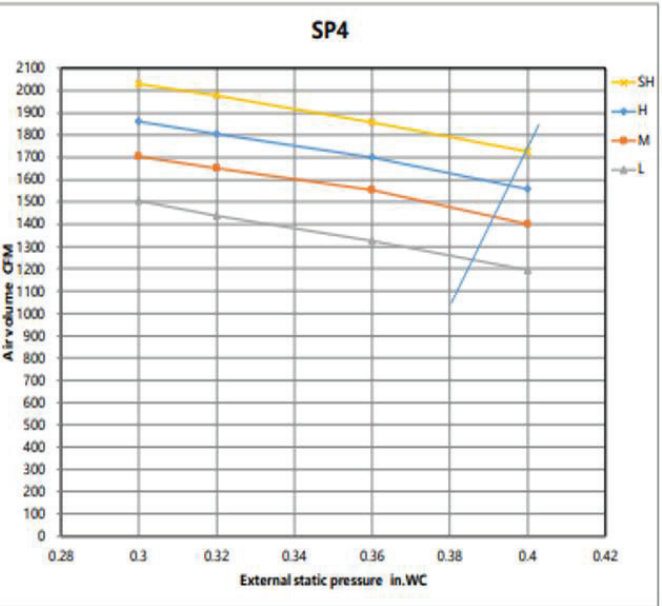
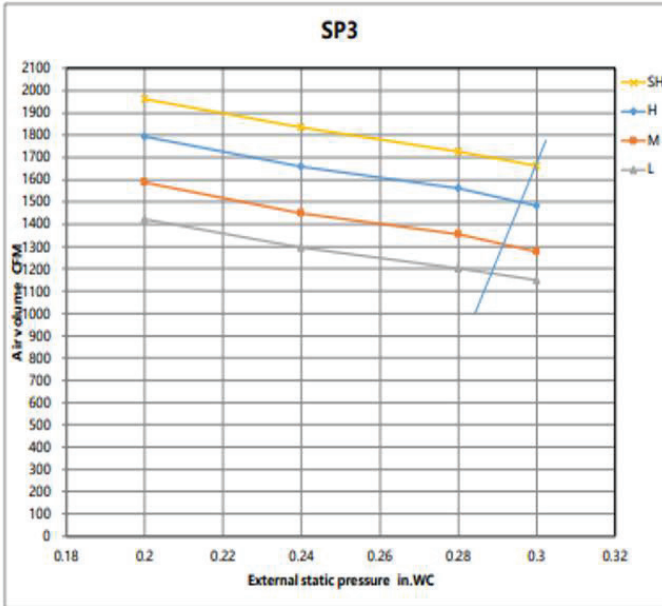
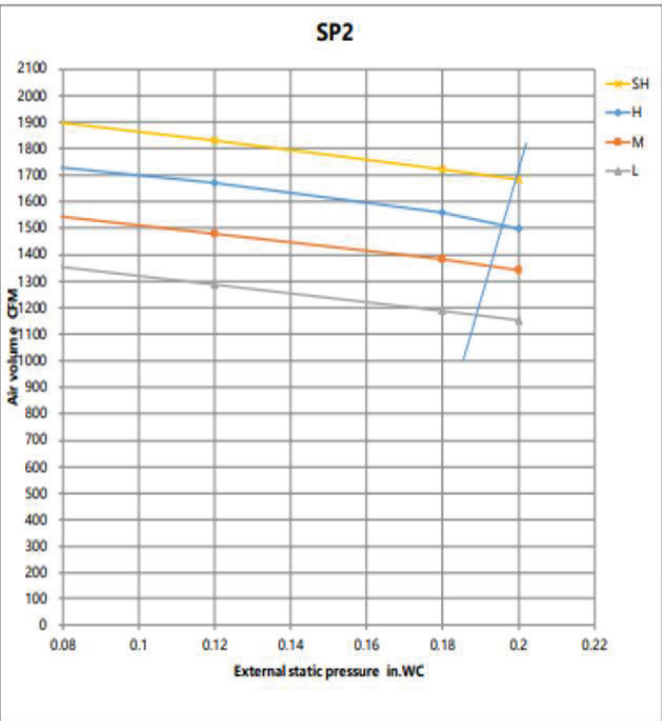
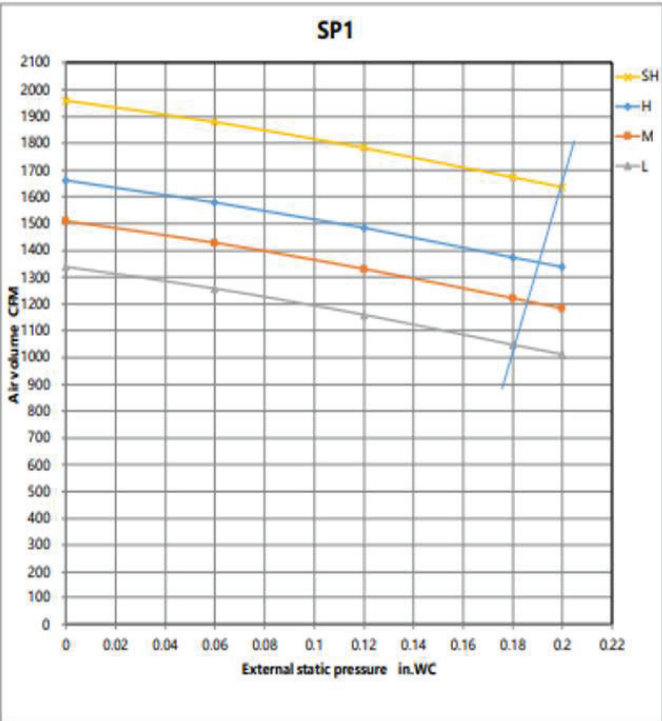


Outdoor Unit Dimensions and Connections

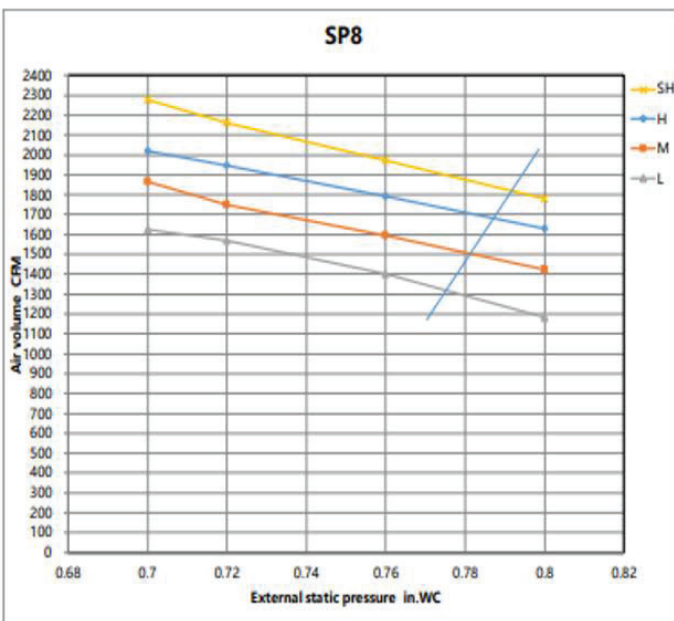
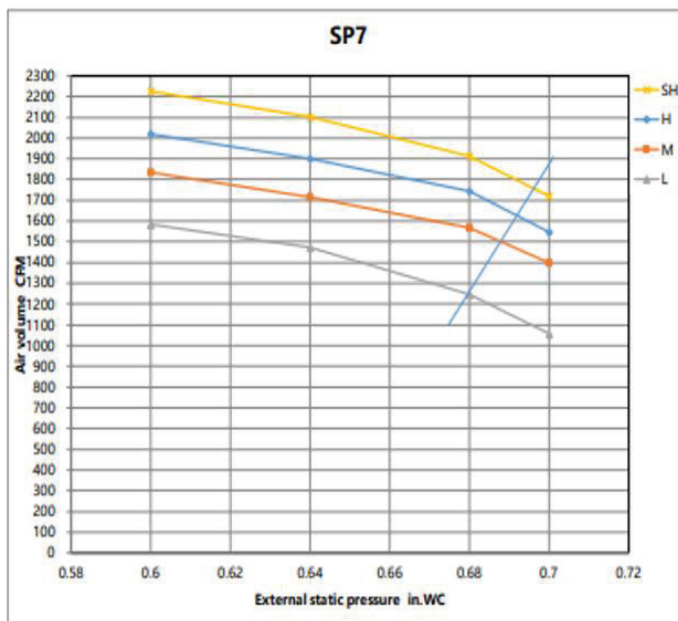
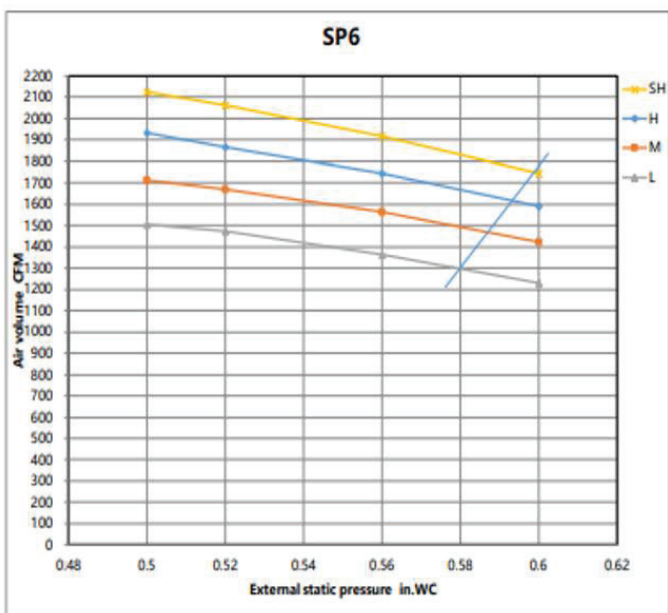
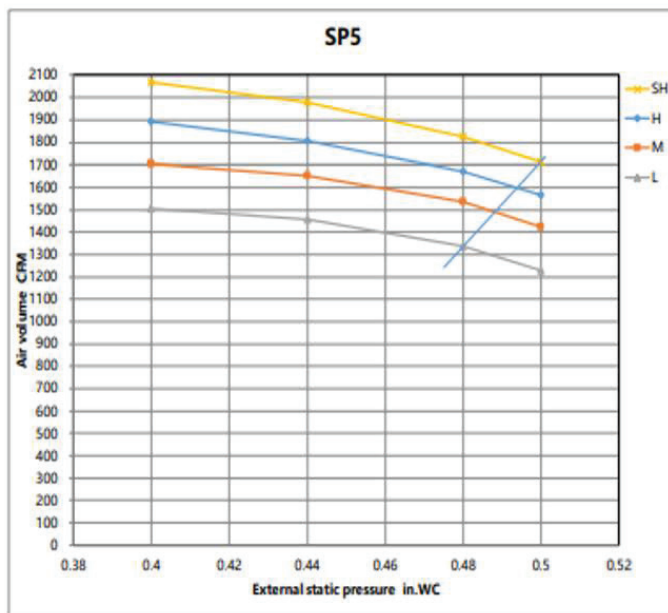


Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)
BMS500-AAS048-1CSXLD	37.48	16.34	52.48	41.0	37.48	25.0	15.9	17.6

Air Volume

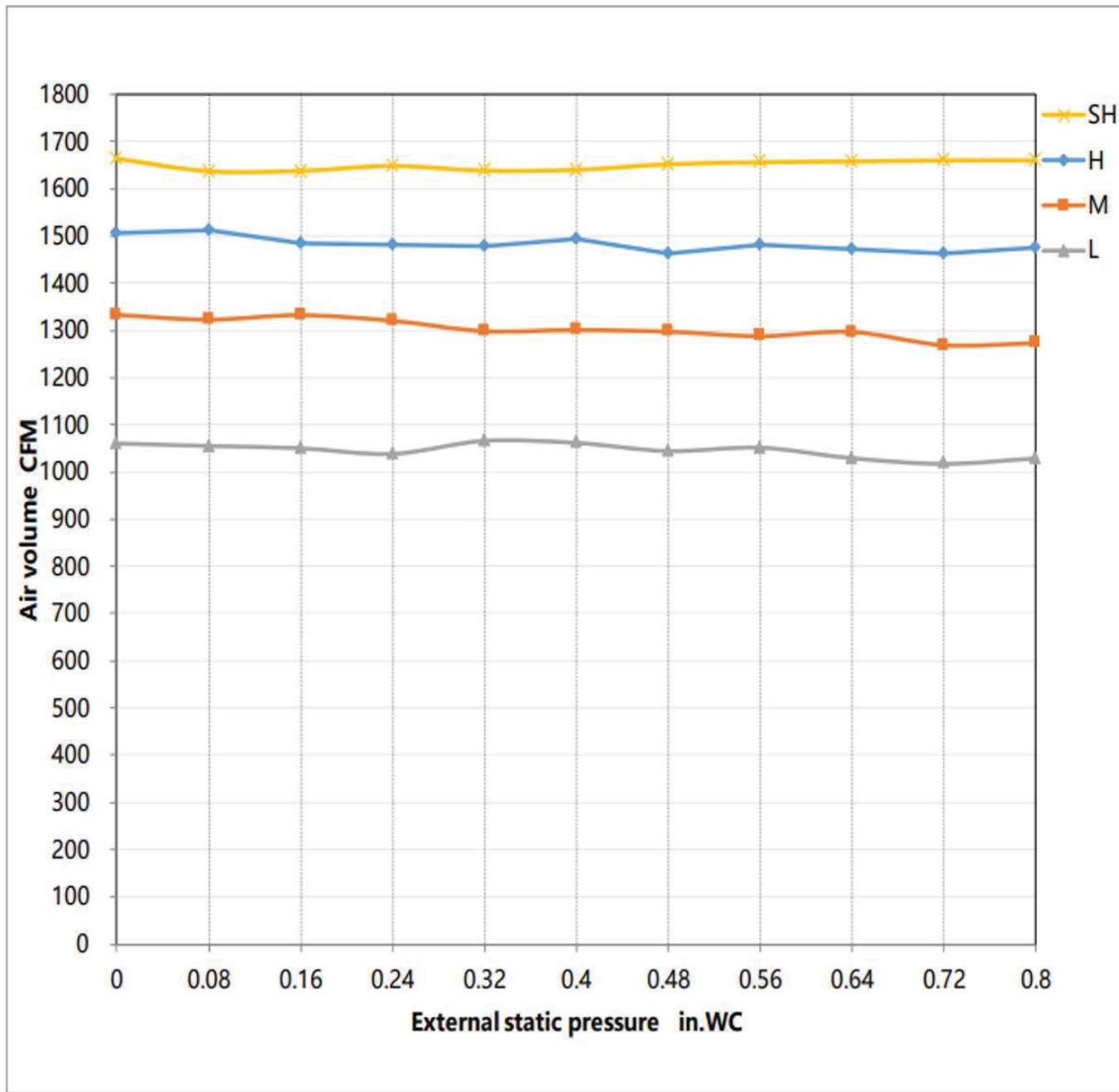


Air Volume



Air Volume

Constant air volume



Climate 5000 Series - Gen 4

48K Btu Single Zone Regular Light Commercial System - Ducted IDU


BOSCH

Cooling Performance - Indoor Dry Bulb Temperature at 1647 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F (16.0°C)					64.4°F (18.0°C)					67°F (19.4°C)					71.6°F (22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	53673	53649	54414	54640	56143	56000	55954	55919	55913	55867	57675	57602	57547	57539	57484	60836	60819	60694	60671	60543
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	3.16	3.16	3.16	3.16	3.15	3.14	3.15	3.15	3.15	3.15	3.14	3.14	3.14	3.14	3.14	3.16	3.16	3.16	3.16	3.16
-4°F (-20.0°C)	TC	57802	57775	58600	58843	60462	60308	60258	60220	60215	60165	62111	62033	61974	61966	61906	65515	65497	65363	65338	65200
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	3.28	3.28	3.27	3.27	3.26	3.26	3.26	3.26	3.26	3.26	3.25	3.25	3.26	3.26	3.26	3.27	3.27	3.27	3.27	3.28
0°F (-17.8°C)	TC	59720	59685	60793	61046	62723	62282	62213	62165	62157	62782	64112	64021	63942	63932	63902	67560	67544	67539	67548	67592
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	3.30	3.30	3.30	3.30	3.30	3.30	3.30	3.30	3.30	3.30	3.30	3.30	3.31	3.31	3.31	3.34	3.34	3.34	3.34	3.34
5°F (-15.0°C)	TC	61432	61644	63053	63344	65296	64314	64330	64334	64400	65360	66370	66387	66401	66403	66491	70263	70242	70279	70286	70317
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	3.33	3.33	3.34	3.34	3.35	3.34	3.34	3.34	3.35	3.36	3.37	3.37	3.37	3.37	3.37	3.44	3.44	3.44	3.44	3.44
14°F (-10.0°C)	TC	64637	64605	65924	66233	68287	67846	67787	67740	67733	68367	70171	70083	70012	70001	69930	74614	74591	74448	74422	74263
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	3.79	3.79	3.78	3.78	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.77	3.78	3.78	3.79	3.79	3.79
17°F (-8.3°C)	TC	65549	65763	67282	67611	69807	68812	68840	68853	68911	69882	71160	71187	71215	71218	71306	75681	75660	75648	75662	75726
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.90	3.91	3.93	3.93	3.94	3.94	3.94
23°F (-5.0°C)	TC	62919	63190	64824	65145	67291	66105	66125	66132	66219	67364	68385	68409	68427	68428	68539	72714	72692	72743	72753	72795
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	3.74	3.74	3.75	3.75	3.78	3.76	3.76	3.76	3.76	3.78	3.79	3.79	3.79	3.79	3.79	3.86	3.86	3.86	3.86	3.86
35°F (1.7°C)	TC	58221	58223	58565	58728	60623	60942	60940	60945	60945	61108	62850	62846	62849	62851	62857	66389	66373	66370	66367	66373
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	3.22	3.22	3.24	3.25	3.39	3.41	3.41	3.41	3.41	3.42	3.56	3.56	3.56	3.56	3.56	3.56	3.57	3.57	3.57	3.57
41°F (5.0°C)	TC	56411	56411	56860	57256	58948	59034	59033	59036	59036	59243	60876	60872	60875	60876	60880	64291	64276	64272	64269	64276
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	2.93	2.93	2.96	2.99	3.11	3.12	3.12	3.12	3.12	3.14	3.26	3.26	3.26	3.26	3.26	3.25	3.55	3.55	3.55	3.55
47°F (8.3°C)	TC	54597	54633	55366	55613	57259	57128	57126	57128	57127	57392	58903	58897	58901	58901	58901	62200	62185	62179	62177	62183
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	2.66	2.66	2.71	2.73	2.85	2.84	2.84	2.84	2.84	2.86	2.97	2.97	2.97	2.97	2.97	3.24	3.24	3.23	3.23	3.23
50°F (10.0°C)	TC	53296	54763	56611	56939	59147	56305	56336	56725	56904	59203	58501	58577	58608	58611	59175	62533	62698	62860	62879	62957
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	3.08	3.08	3.08	3.08	3.07	3.08	3.08	3.08	3.08	3.07	3.07	3.07	3.07	3.07	3.07	3.07	3.07	3.07	3.07	3.07
59°F (15.0°C)	TC	52328	54164	56086	56427	58723	55365	55441	56070	56488	58781	57649	57708	57728	57812	58830	61904	62028	62151	62165	62217
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	3.37	3.38	3.38	3.39	3.40	3.38	3.38	3.39	3.39	3.40	3.39	3.39	3.39	3.39	3.40	3.42	3.41	3.41	3.41	3.41
68°F (20.0°C)	TC	51341	53533	55518	55871	58242	54353	54475	55578	55931	58300	56703	56745	56847	56897	58347	61097	61211	61307	61316	61348
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	3.66	3.68	3.70	3.70	3.72	3.69	3.69	3.70	3.70	3.72	3.71	3.71	3.71	3.71	3.72	3.76	3.76	3.76	3.76	3.76
77°F (25.0°C)	TC	50320	52743	54807	55172	57637	53346	53515	54867	55234	57695	55762	55824	55967	56029	57744	60225	60408	60547	60562	60619
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	3.98	4.00	4.02	4.02	4.04	4.01	4.01	4.02	4.02	4.04	4.03	4.03	4.03	4.03	4.04	4.08	4.07	4.07	4.07	4.07
82°F (27.8°C)	TC	50055	52610	54732	55103	57627	53043	53247	54780	55165	57687	55510	55580	55753	55822	57738	60018	60244	60401	60417	60483
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	4.31	4.33	4.35	4.35	4.37	4.34	4.34	4.35	4.35	4.37	4.36	4.36	4.36	4.36	4.37	4.41	4.41	4.40	4.40	4.40
86°F (30.0°C)	TC	50392	52908	55071	55455	58038	53143	53387	55132	55519	58098	55661	55725	55929	56010	58148	60259	60492	60635	60650	60707
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	4.76	4.78	4.81	4.81	4.84	4.79	4.79	4.81	4.81	4.84	4.82	4.82	4.82	4.82	4.84	4.88	4.88	4.87	4.87	4.87
95°F (35.0°C)	TC	50775	51049	52699	53011	55106	53963	53995	54012	54075	55176	56279	56291	56325	56330	56424	60643	60620	60632	60650	60731
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	5.55	5.56	5.56	5.56	5.78	5.72	5.72	5.72	5.72	5.78	5.85	5.85	5.85	5.85	5.86	6.09	6.09	6.09	6.09	6.09
104°F (40.0°C)	TC	43242	43638	45281	45553	47370	45966	45992	46074	46123	47429	47928									

Climate 5000 Series - Gen 4

48K Btu Single Zone Regular Light Commercial System - Ducted IDU



Heating Performance - Indoor Dry Bulb Temperature at 1647 CFM										
Outdoor DB(°F)	TCITotal Heating Capacity(BTU/hr)					PIPower Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	28465	27896	27757	27646	27397	4.71	4.75	4.76	4.77	4.79
-4°F(-20.0°C)	33728	33054	32888	32757	32462	4.99	5.03	5.04	5.05	5.07
0°F(-17.8°C)	36677	35943	35763	35620	35300	5.15	5.20	5.21	5.22	5.24
5°F(-15.0°C)	40362	39555	39357	39200	38847	5.36	5.40	5.41	5.42	5.45
14°F(-10.0°C)	45748	44833	44608	44430	44030	5.64	5.68	5.69	5.70	5.73
17°F(-8.3°C)	47543	46592	46359	46173	45758	5.73	5.77	5.79	5.80	5.82
22°F(-5.6°C)	50437	49428	49181	48985	48544	5.76	5.81	5.82	5.83	5.85
27°F(-2.8°C)	53332	52265	52004	51796	51330	5.79	5.84	5.85	5.86	5.89
32°F(0.0°C)	56226	55102	54826	54607	54116	5.83	5.87	5.88	5.90	5.92
35°F(1.7°C)	57963	56804	56520	56294	55787	5.84	5.89	5.90	5.92	5.94
37°F(2.8°C)	58616	57443	57156	56927	56415	5.71	5.76	5.77	5.78	5.81
42°F(5.6°C)	60247	59042	58747	58512	57985	5.38	5.43	5.44	5.45	5.47
44.6°F(7.0°C)	60900	59682	59383	59146	58613	5.25	5.29	5.30	5.31	5.34
47°F(8.3°C)	61879	60641	60338	60096	59556	5.05	5.09	5.10	5.11	5.13
52°F(11.1°C)	61057	59836	59537	59299	58765	4.62	4.66	4.67	4.68	4.70
57°F(13.9°C)	60236	59031	58736	58501	57974	4.19	4.22	4.23	4.24	4.26
59°F(15.0°C)	59907	58709	58415	58182	57658	4.01	4.05	4.06	4.06	4.08
62°F(16.7°C)	59414	58226	57935	57703	57184	3.75	3.79	3.79	3.80	3.82
64.4°F(18.0°C)	59020	57839	57550	57320	56804	3.55	3.58	3.58	3.59	3.60
69.8°F(21.0°C)	52462	51413	51156	50951	50493	3.08	3.10	3.11	3.12	3.13
75.2°F(24.0°C)	45905	44987	44762	44583	44182	2.61	2.63	2.64	2.64	2.65

Bolded values were tested. All other values are simulated and for reference only.